Mini Project (15 marks):

1. **Create a group of 3 students** *(Preferably from same batch).*
2. **Identify a real-world problem relevant to any application domain such as:**
   - Healthcare
   - Agriculture
   - Finance
   - Education
   - Transportation
   - Smart Cities
   - E-commerce
   - Environmental Monitoring
   - Cybersecurity
   - Social Media Analytics etc
3. **Collect, clean and split the suitable dataset for your problem statement**
4. **Compare with the State-of-the-art Models with performance metrics**
5. **Develop a simple and interactive user interface using any suitable framework such as:**
   - Streamlit
   - Gradio
   - Flask
   - Django etc
6. **Maintain code quality and upload clean code on GitHub repo.**
7. **Prepare a comprehensive project report and submit on GC.**

   Report structure:

   1. Abstract
   2. Introduction
   3. Literature Review *(At least 5 latest papers should be discussed and compared)*
   4. Methodology
   5. Data Cleaning and preprocessing
   6. Results and discussion
      i. State-of the-art models comparison *(at least 4 models)* with performance metrics (tables and neat & clean figures)
      ii. User Interface Development and Deployment
   7. References

Note:

- A demo and viva have been scheduled in which students have to justify above metrics and put all points in the **10** pages report.
- Your problem statement and dataset should not match with any other groups.
- Every project must be original and plagiarism free.
- All team members should be able to explain the complete workflow during the project demonstration.
- Projects should demonstrate understanding of the Machine Learning pipeline rather than only achieving high accuracy.